RELATIVE PSEUDO-COHERENCE IS PRESERVED BY DERIVED BASE CHANGE ALONG A RELATIVELY PSEUDO-COHERENT ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 21567-21576 (canonical tag 067B) and the counted proof is the proof environment at source lines 21578-21619, which immediately follows it and is its own proof.

Editorial note. The proof refers three times to earlier lemmas of the same chapter, namely the quasi-isomorphism for derived Tor, the cone criterion for relative pseudo-coherence and the base change lemma; all three are carried as navigation stubs with their roles recorded in the reasons.

Editorial note. The author writes the polynomial variables as y_1, ..., y_m in the presentations and as y_1, ..., y_n once in the resolution, and the derived category statement is written with the symbol for the total complex; both are copied exactly as printed.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- A relatively pseudo-coherent algebra admits a resolution by a complex of finite free modules over a polynomial algebra, and a bounded above complex may be replaced by one of flat modules when testing relative pseudo-coherence. used at proof lines 21579-21585. Produces the resolution E and the flat representative of K that the displayed computation uses.
- Tensoring with a bounded above complex of flat modules computes the derived tensor product, and replacing the second factor by a quasi-isomorphic complex of flat modules does not change the total complex up to quasi-isomorphism. used at proof lines 21586-21601. Justifies the chain of equalities and the quasi-isomorphism in the display that rewrites the derived base change as a total complex over R.

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- A total complex filtered by subcomplexes whose successive quotients are m-pseudo-coherent is m-pseudo-coherent, and the cone criterion propagates relative pseudo-coherence along such filtrations. used at proof lines 21605-21612. Reduces the assertion for the total complex to the assertions for the single filtration quotients.

Source reference. Original lemma 064K: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14284–14284. The reference supplies the quasi-isomorphism that replaces the second factor by the finite free resolution E in the total complex, used at proof lines 21599-21601; the display and the filtration argument that follow are inside the counted span.

Source reference. Original lemma 0674: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21393–21393. The reference is the cone criterion invoked at proof lines 21610-21612 to pass from the filtration quotients to the total complex; the filtration and the reduction to a single quotient are inside the counted span.

Source reference. Original lemma 067A: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21543–21543. The reference is the base change statement for a polynomial algebra extension, invoked at proof lines 21616-21618 as the last remaining case; the reduction of the general case to that one is inside the counted span.

Lemma 0.2. *Let $R \to A \to B$ be finite type ring maps. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a complex of A-modules. Assume B as a B-module is pseudo-coherent relative to A. If $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R, then $K^\bullet \otimes^{\mathbf{L}}_A B$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R.*

Proof. Choose a surjection $A[y_1, \ldots, y_m] \to B$. Choose a surjection $R[x_1, \ldots, x_n] \to A$. Combined we get a surjection $R[x_1, \ldots, x_n, y_1, \ldots y_m] \to B$. Choose a resolution $E^\bullet \to B$ of B by a complex of finite free $A[y_1, \ldots, y_n]$-modules (which is possible by our assumption on the ring map $A \to B$). We may assume that $K^\bullet$ is a bounded above complex of flat A-modules. Then

$$\begin{aligned} K^\bullet \otimes^{\mathbf{L}}_A B &= \mathrm{Tot}(K^\bullet \otimes_A B[0]) \\ &= \mathrm{Tot}(K^\bullet \otimes_A A[y_1, \ldots, y_m] \otimes_{A[y_1,\ldots,y_m]} B[0]) \\ &\cong \mathrm{Tot}\left((K^\bullet \otimes_A A[y_1, \ldots, y_m]) \otimes_{A[y_1,\ldots,y_m]} E^\bullet\right) \\ &= \mathrm{Tot}(K^\bullet \otimes_A E^\bullet) \end{aligned}$$

in $D(A[y_1, \ldots, y_m])$. The quasi-isomorphism $\cong$ comes from an application of Lemma 064K. Thus we have to show that $\mathrm{Tot}(K^\bullet \otimes_A E^\bullet)$ is m-pseudo-coherent as a complex of $R[x_1, \ldots, x_n, y_1, \ldots y_m]$-modules. Note that $\mathrm{Tot}(K^\bullet \otimes_A E^\bullet)$ has a filtration by subcomplexes with successive quotients the complexes $K^\bullet \otimes_A E^i[-i]$. Note that for $i \ll 0$ the complexes $K^\bullet \otimes_A E^i[-i]$ have zero cohomology in degrees $\leq m$ and hence are m-pseudo-coherent (over any ring). Hence, applying Lemma 0674 and induction, it suffices to show that $K^\bullet \otimes_A E^i[-i]$ is pseudo-coherent relative to R for all i. Note that $E^i = 0$ for $i > 0$. Since also E^i is finite free this reduces to proving that $K^\bullet \otimes_A A[y_1, \ldots, y_m]$ is m-pseudo-coherent relative to R which follows from Lemma 067A for instance. □